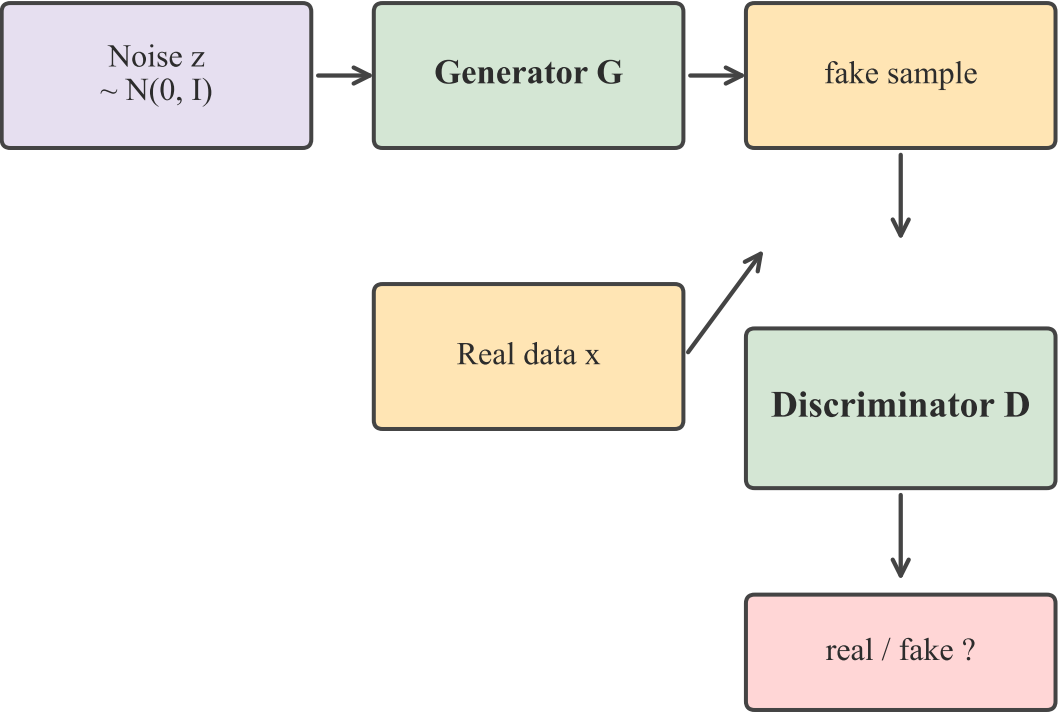


Standard GAN vs CAARMA — D's role is conceptually identical

Standard GAN (Goodfellow 2014)



G: fool D into saying "real"

D: correctly label real vs fake

→ D is discarded after training

CAARMA mapping

(same alternating game, different inputs)

z (random noise)	→ real audio waveform
G (generator)	→ MFA-Conformer encoder M
G's output	→ e (encoder embedding)
(no analog in GAN)	→ + e_syn = mixup(e_i, e_j)
D	→ MixupDiscriminator (HuBERT-backed)
D's job	→ tell real e from synthetic e_syn

→ D is also discarded at inference (same as standard GAN)